
MODEL SELECTION AND VALIDATION REPORT (remove_7_biomass model)

OBSERVATIONS:1690

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_7_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.54	0.18
remove_7_kde_fric	-0.07	0.04
remove_7_kde_feve	-0.27	0.03
sbt_mean	0.21	0.10
sbt_sd	0.05	0.05
sst_sd	-0.03	0.05
log_chla_mean	0.06	0.04
depth	0.02	0.05
log_sum_fishing	0.06	0.04

Dispersion parameter: 1.05

Spatiotemporal AR1 correlation (rho): 0.90

Matérn range: 195.03

Spatiotemporal marginal AR1 SD: 0.94

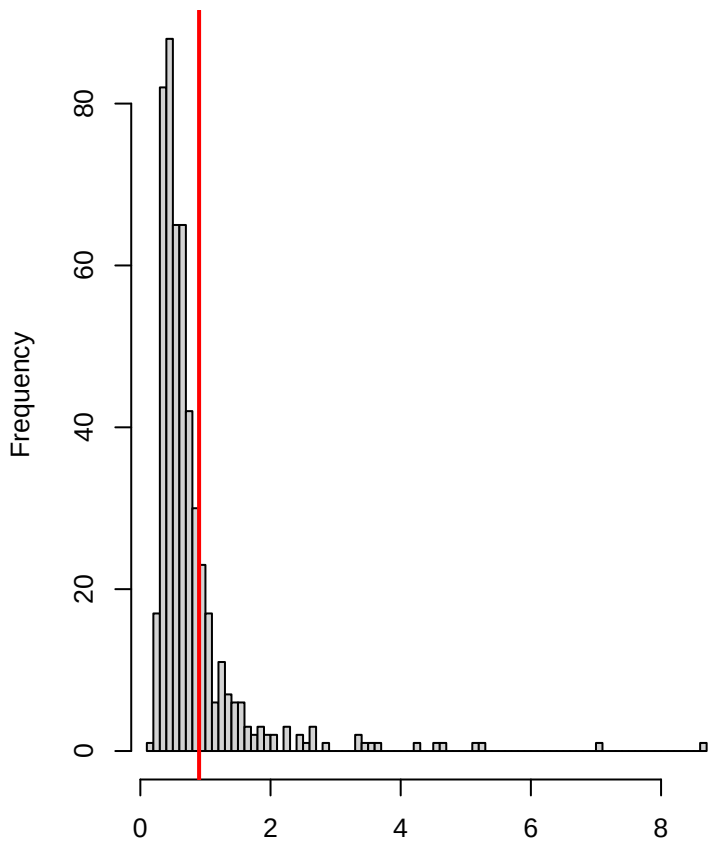
REML criterion at convergence: 10940.973

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

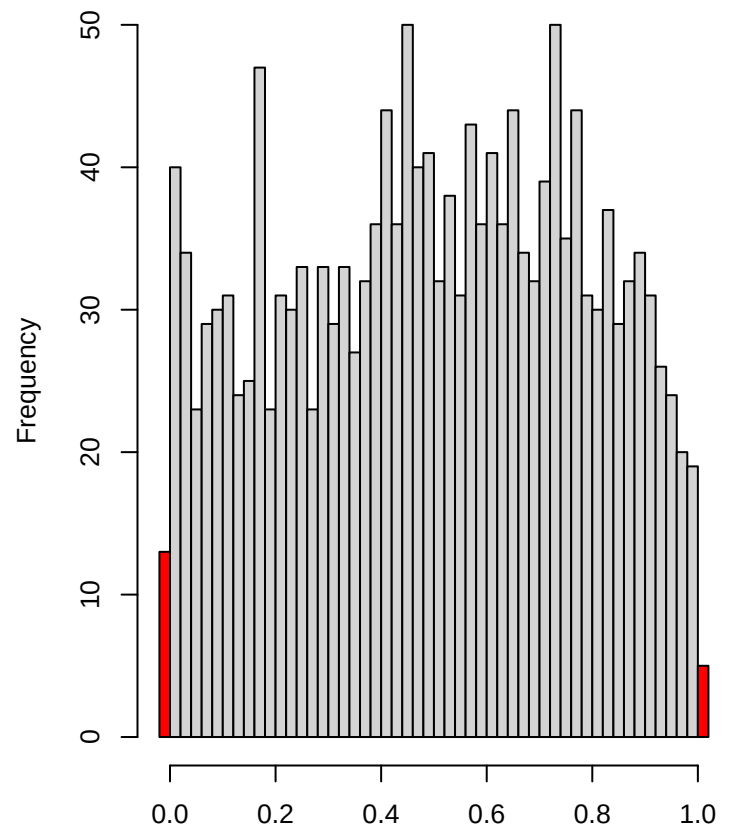
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	21907.95	1.093197
spatiotemporal_random_fields_ar1_with_anisotropy	15	21909.04	75.615405
spatial_random_fields	12	21984.65	3.377403
spatial_random_fields_with_anisotropy	14	21988.03	94.050772
spatiotemporal_random_fields_iid_with_anisotropy	14	22082.08	3.977368
spatiotemporal_random_fields_iid	12	22086.06	126.779915
error_only	10	22212.84	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated

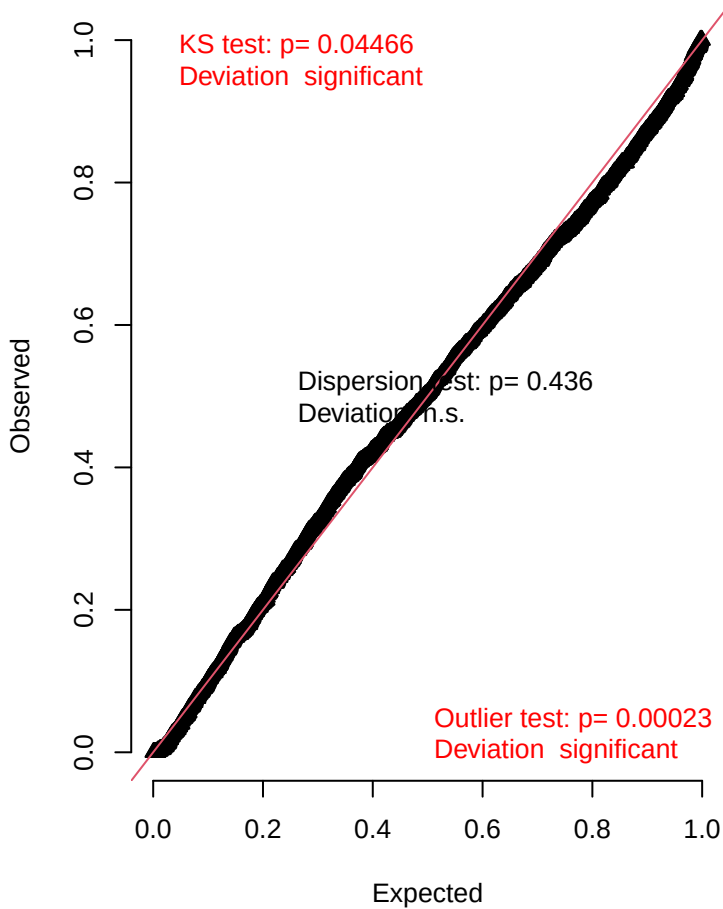


Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

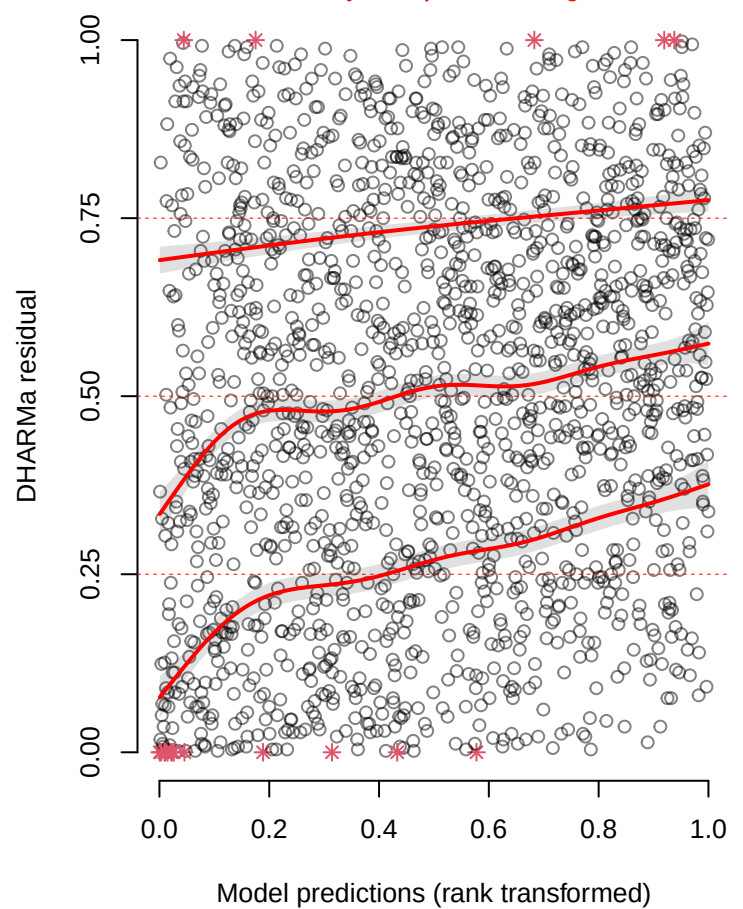


QQ plot residuals



Residuals (outliers are marked red)

DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.021 | H1: Distance-based autocorrelation | p-value = 0.019

